Unit-6

Machine Learning Using Python

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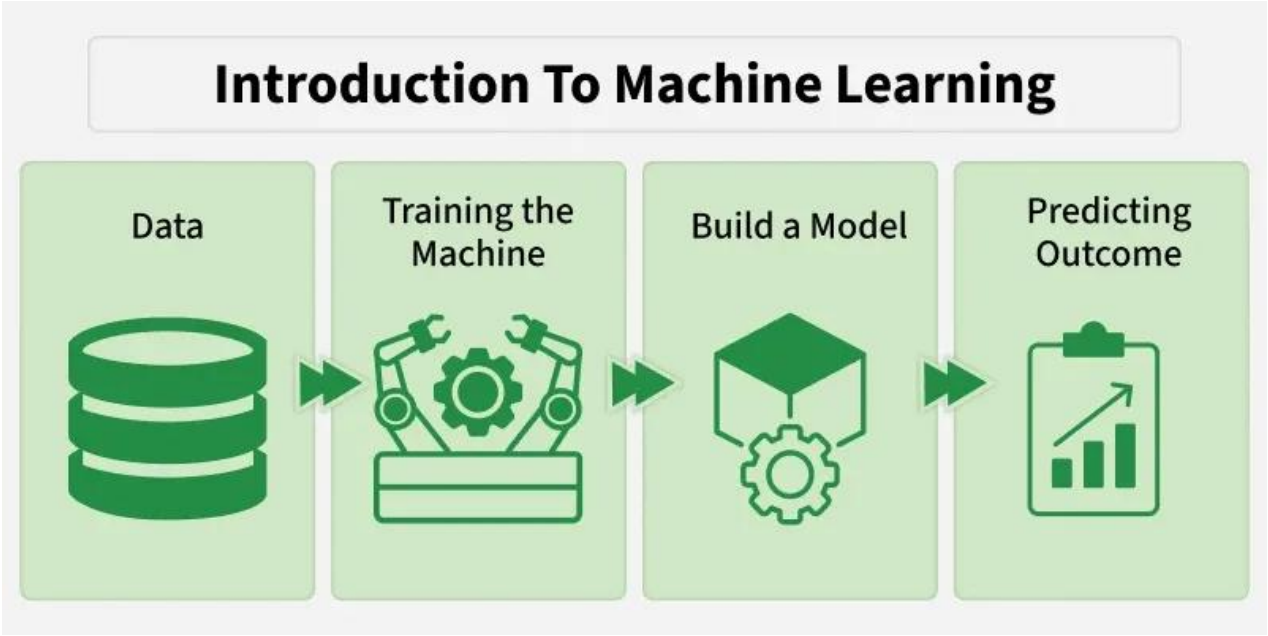
Contents

- Introduction to Machine Learning
- Supervised Vs Unsupervised Learning
- Python Libraries for Machine Learning
- Regression
- Classification
- Unsupervised Learning
- Recommender System

Introduction

- Machine learning (ML) allows computers to learn and make decisions without being explicitly programmed.
- It involves feeding data into algorithms to identify patterns and make predictions on new data. It is used in various applications like image recognition, speech processing, language translation, recommender systems, etc.



Why do we need Machine Learning?

- Traditional programming requires exact instructions and doesn't handle complex tasks like understanding images or language well. It can't efficiently process large amounts of data. Machine Learning solves these problems by learning from examples and making predictions without fixed rules. Let's see various reasons why it is important:

1. Solving Complex Business Problems

- Traditional programming struggles with tasks like language understanding and medical diagnosis. ML learns from data and predicts outcomes easily.

Examples:

- Image and speech recognition in healthcare.
- Language translation and sentiment analysis.

2. Handling Large Volumes of Data

- The internet generates huge amounts of data every day. Machine Learning processes and analyzes this data quickly by providing valuable insights and real-time predictions.

Examples:

- Fraud detection in financial transactions.
- Personalized feed recommendations on Facebook and Instagram from billions of interactions.

3. Automate Repetitive Tasks

- ML automates time-consuming, repetitive tasks with high accuracy hence reducing manual work and errors.

Examples:

- Gmail filtering spam emails automatically.
- Chatbots handling order tracking and password resets.
- Automating large-scale invoice analysis for key insights.

4. Personalized User Experience

- ML enhances user experience by tailoring recommendations to individual preferences. It analyze user behavior to deliver highly relevant content.

Examples:

- Netflix suggesting movies and TV shows based on our viewing history.
- E-commerce sites recommending products we're likely to buy.

5. Self Improvement in Performance

- ML models evolve and improve with more data helps in making them smarter over time. They adapt to user behavior and increase their performance.

Examples:

- Voice assistants like Siri and Alexa learning our preferences and accents.
- Search engines refining results based on user interaction.
- Self-driving cars improving decisions using millions of miles of driving data.

What Makes a Machine "Learn"?

- A machine "learns" by identifying patterns in data and improving its ability to perform specific tasks without being explicitly programmed for every scenario. This learning process helps machines to make accurate predictions or decisions based on the information they receive. Unlike traditional programming where instructions are fixed, ML allows models to adapt and improve through experience.

Here is how the learning process works:

- **Data Input:** Machine needs data like text, images or numbers to analyze. Good quality and enough quantity of data are important for effective learning.
- **Algorithms:** Algorithms are mathematical methods that help the machine find patterns in data. Different algorithms help different tasks such as classification or regression.
- **Model Training:** During training, the machine adjusts its internal settings to better predict outcomes. It learns by reducing the difference between its predictions and actual results.
- **Feedback Loop:** Machine compares its predictions with true outcomes and uses this feedback to correct errors. Techniques like gradient descent help it update and improve.
- **Experience and Iteration:** Machine repeats training many times with data helps in refining its predictions with each pass, more data and iterations improve accuracy.
- **Evaluation and Generalization:** Model is tested on unseen data to ensure it performs well on real-world tasks.
- Machines "learn" by continuously increasing their understanding through data-driven iterations like how humans learn from experience.

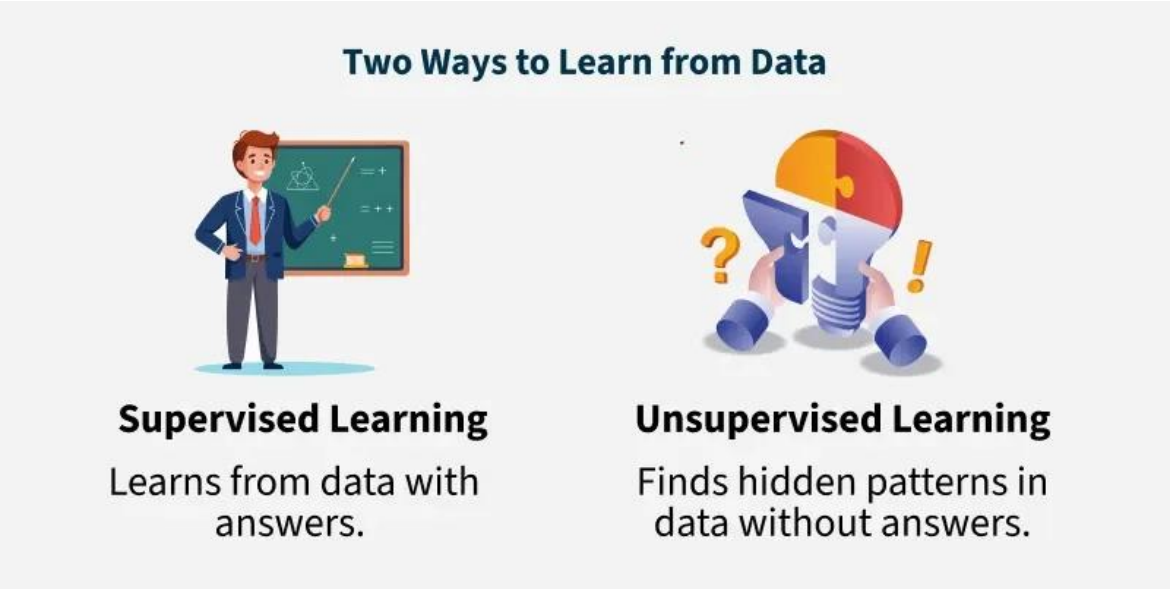
Importance of Data in Machine Learning

- Data is the foundation of machine learning (ML) without quality data ML models cannot learn, perform or make accurate predictions.
- Data provides the examples from which models learn patterns and relationships.
- High-quality and diverse data improves how well models perform and generalize to new situations.
- It helps models to understand real-world scenarios and adapt to practical uses.
- Features extracted from data are important for effective training.
- Separate datasets for validation and testing measure how well the model works on unseen data.
- Data drives continuous improvements in models through feedback loops.

Types of Machine Learning

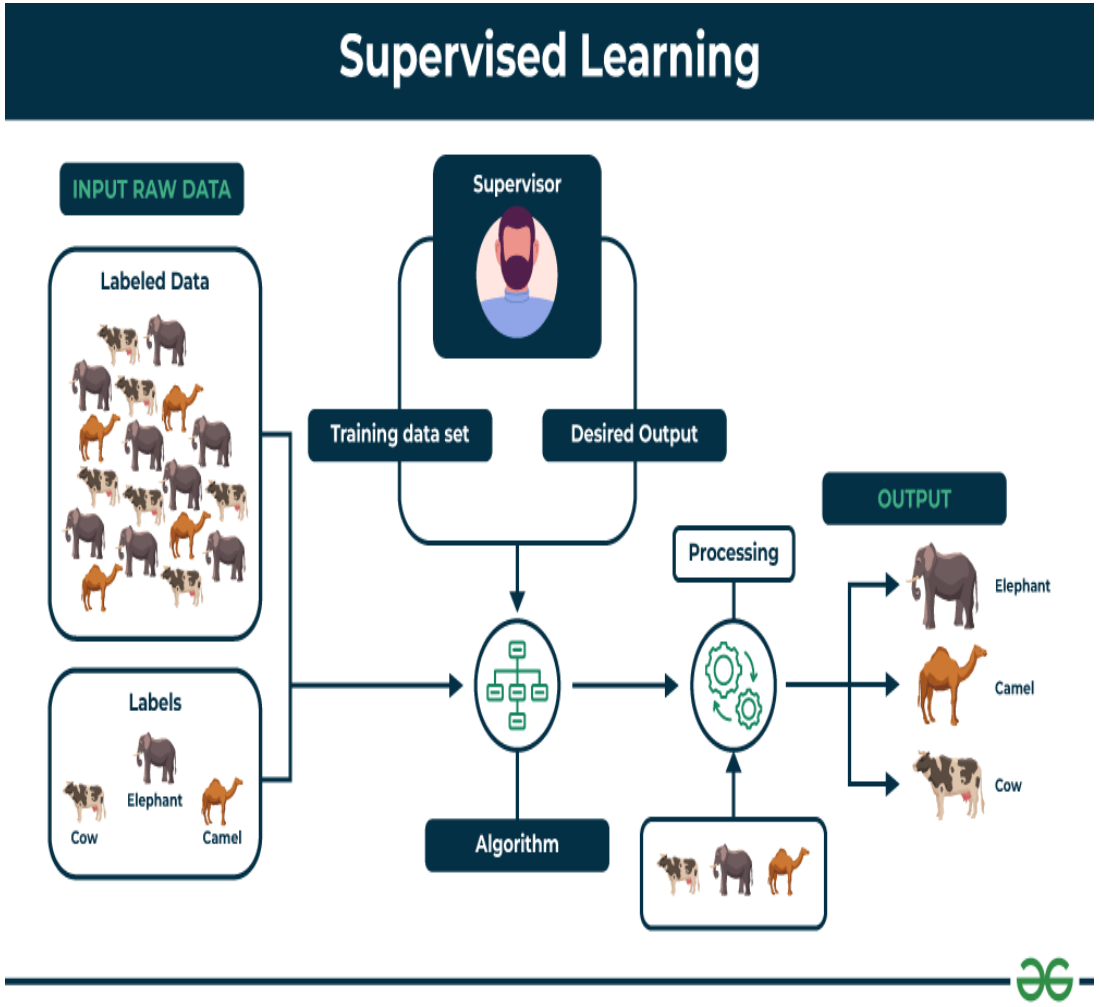
There are three main types of machine learning which are as follows:

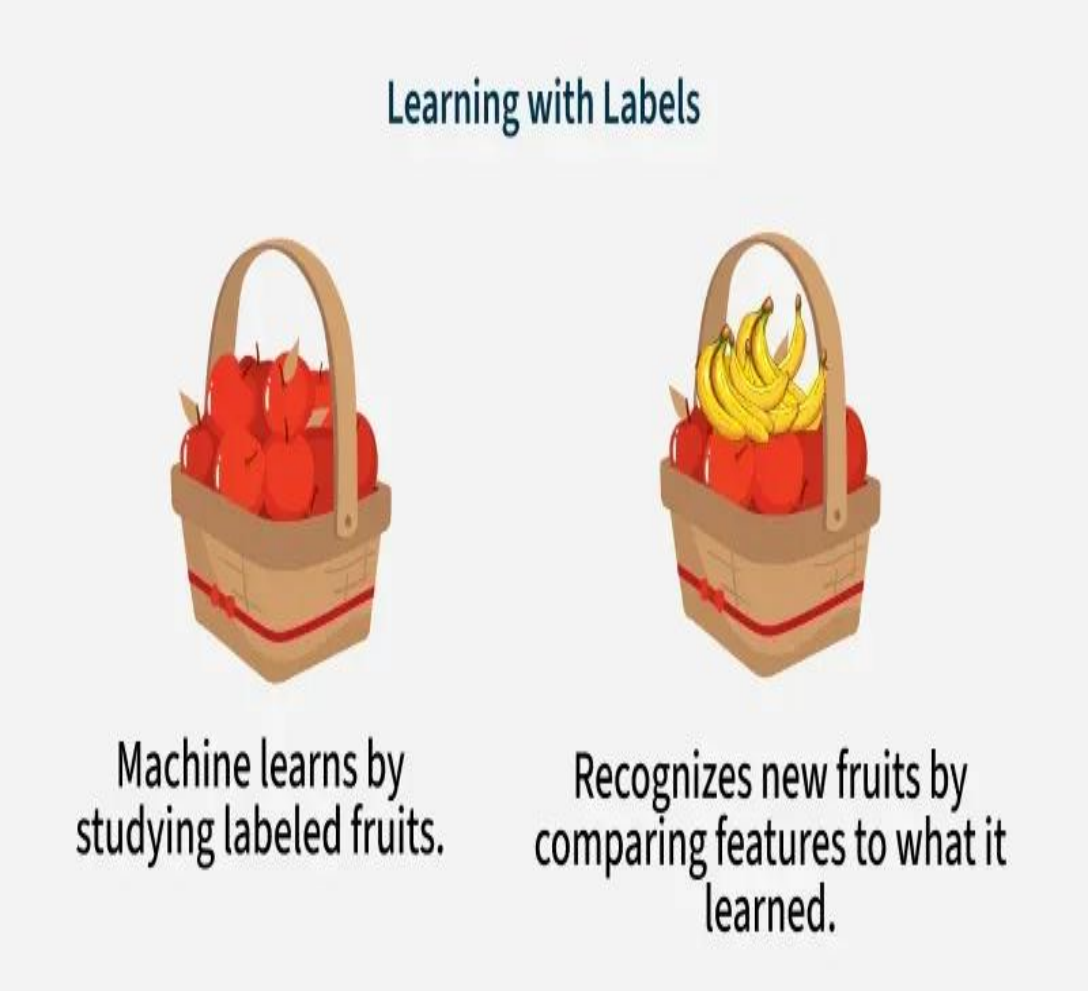
- **Supervised Learning**
- **Unsupervised Learning**
- **Reinforcement Learning**



Supervised Learning

- Supervised learning as the name suggests, works like a **teacher** or **supervisor** guiding the machine. In this approach we teach or train the machine using the labelled data(correct answers or classifications)
- which means each input has the correct output in the form of answer or category attached to it.
- After that machine is provided with a new set of examples (data) so that it can analyses the training data and produces a correct outcome from labeled data.
- **For example**, a labeled dataset of images of Elephant, Camel and Cow would have each image tagged with either "**Elephant**", "**Camel**" or "**Cow**."


Supervised Learning
INPUT RAW DATA
Supervisor
Labeled Data
Training data set
Desired Output
OUTPUT
Processing
Elephant
Camel
Cow
Labels
Elephant
Cow
Camel
Algorithm


Learning with Labels
Machine learns by studying labeled fruits.
Recognizes new fruits by comparing features to what it learned.

Example to Understand

- Imagine we have a basket full of different fruits that we want the machine to identify. The machine first looks at the image of a fruit and extracts features like its shape, color and texture. Then it compares these features to the fruits it has already learned during training. If the new fruit's features closely match those of an apple, the machine will predict that the fruit is an apple.
- For example, suppose we train the machine by showing it fruits one by one:
- If the fruit is round, has a small depression at the top and is red, it is labeled as an Apple.
- If the fruit is long, curved and greenish-yellow, it is labeled as a Banana.
- Now after this training, if we give the machine a new fruit (say a banana) from the basket and ask it to identify it, the machine will use what it has learned during training. It will analyze the shape and color of the new fruit and classify it as a Banana placing it in the correct category. In this way, the machine learns from the training data (the basket with labeled fruits) and applies that knowledge to recognize new, unseen fruits.

Types of Supervised Learning

- Supervised learning is classified into two types of algorithms:

1. Regression

- A regression is used to predict **continuous** values such as house prices, stock prices or temperature. Regression algorithms learn how to connect input data to a specific number or value.
- Some common regression algorithms include:
- Linear Regression
- Polynomial Regression
- Lasso Regression
- Ridge Regression

2. Classification

- A classification is used to predict **categorical** values such as whether a customer will buy or not, whether an email is spam or not or whether a medical image shows a tumor or not. Classification algorithms learn how to connect input data to the probability of belonging to different groups or categories.
- Some of the most common classification algorithms include:
- Logistic Regression
- Support Vector Machines
- Decision Trees
- Random Forests
- Naive Baye

Regression

- A regression is used to predict **continuous** values such as house prices, stock prices or temperature. Regression algorithms learn how to connect input data to a specific number or value.

Types of Regression

- Regression can be classified into different types based on the number of predictor variables and the nature of the relationship between variables

1. Simple Linear Regression

Simple Linear Regression models the relationship between one independent variable and a continuous dependent variable by fitting a straight line that minimizes the sum of squared errors. It assumes a constant rate of change, meaning the output varies proportionally with the input.

•**Application:** Estimating house price from only its size

•**Advantage:** Highly interpretable due to its simple mathematical structure

•**Disadvantage:** Cannot capture curved or complex data patterns

2. Multiple Linear Regression

Multiple Linear Regression extends simple linear regression by incorporating multiple independent variables to predict a continuous outcome. Each predictor is assigned a coefficient that reflects its individual impact while holding other variables constant.

•**Application:** Predicting house prices using multiple factors like size, location, age and number of rooms

•**Advantage:** Captures the combined influence of many factors simultaneously

•**Disadvantage:** Performance drops in the presence of multicollinearity (features highly correlated with each other)

3. Polynomial Regression

Polynomial Regression models non-linear relationships by transforming input features into higher-degree polynomial terms (e.g x^2, x^3). Although it fits non-linear curves it remains a linear model in terms of parameters.

•**Application:** Modelling curved growth trends like population increase or temperature variation

•**Advantage:** Effectively captures non-linear relationships without switching to non-linear algorithms

•**Disadvantage:** Higher-degree polynomials may lead to overfitting and unstable predictions

4. Ridge and Lasso Regression

Ridge and Lasso are regularized linear regression techniques that add penalty terms to limit large coefficients and reduce overfitting. Ridge (L2) shrinks coefficients smoothly, while Lasso (L1) can reduce some coefficients to zero, enabling feature selection.

•**Application:** Used in high-dimensional datasets like marketing attribution or gene expression data

•**Advantage:** Controls overfitting and improves generalization, especially with many predictors

•**Disadvantage:** Penalty terms make model interpretation less straightforward

5. Support Vector Regression (SVR)

- Support Vector Regression applies the principles of Support Vector Machines to regression tasks. It fits a function within a defined margin (epsilon-tube) and penalizes errors only when predictions fall outside this boundary. Kernel functions allow SVR to model non-linear relationships.
- **Application:** Predicting continuous outcomes such as stock values or energy consumption
- **Advantage:** Works well with high-dimensional, complex datasets and non-linear patterns
- **Disadvantage:** Computationally intensive and requires careful tuning of kernels and parameters

6. Decision Tree Regression

- Decision Tree Regression splits the data into hierarchical branches based on feature thresholds.. Each internal node represents a decision question and leaf nodes represent predicted continuous values. It learns patterns by recursively partitioning the data to minimize prediction errors.
- **Application:** Predicting customer spending behavior based on demographic and financial features
- **Advantage:** Easy to visualize and understand decision logic
- **Disadvantage:** Easily overfits, especially when the tree becomes deep and complex

7. Random Forest Regression

- Random Forest Regression is an ensemble method that builds multiple decision trees using different data samples and averages their predictions. This reduces the overfitting tendency of single trees and improves accuracy through diversity (bagging). Each tree captures a slightly different aspect of the data.
- **Application:** Sales forecasting, demand planning, churn prediction
- **Advantage:** High accuracy and robust performance even on noisy datasets
- **Disadvantage:** Acts as a black-box model, making interpretation difficult due to many trees

Regression Evaluation Metrics

- Evaluation in machine learning measures the performance of a model. Here are some popular evaluation metrics for regression:
- **Mean Absolute Error (MAE):** The average absolute difference between the predicted and actual values of the target variable.
- **Mean Squared Error (MSE):** The average squared difference between the predicted and actual values of the target variable.
- **Root Mean Squared Error (RMSE):** Square root of the mean squared error.
- **Huber Loss:** A hybrid loss function that transitions from MAE to MSE for larger errors, providing balance between robustness and MSE's sensitivity to outliers.
- **R2 – Score:** Higher values indicate better fit ranging from 0 to 1.

Applications

- **Predicting prices:** Used to predict the price of a house based on its size, location and other features.
- **Forecasting trends:** Model to forecast the sales of a product based on historical sales data.
- **Identifying risk factors:** Used to identify risk factors for heart patient based on patient medical data.
- **Making decisions:** It could be used to recommend which stock to buy based on market data.

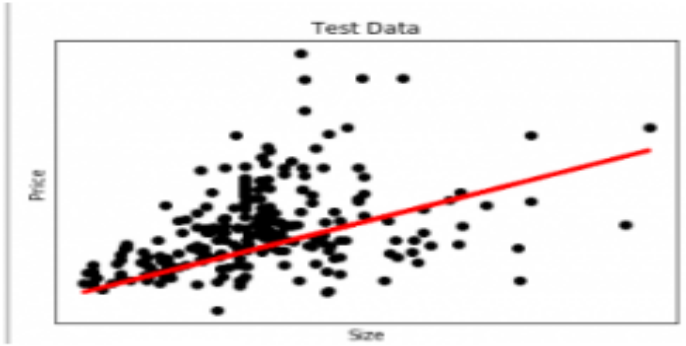
- The housing dataset is loaded with lot size as the input feature and price as the target variable.
- The data is split into training and testing sets and a linear regression model is trained.
- Predictions on test data are plotted against actual values.

```
import pandas as pd
from sklearn import linear_model
import numpy as np
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')
df = pd.read_csv("Housing.csv")
Y = df['price']
X = df['lotsize']
X = X.to_numpy().reshape(len(X), 1)
Y = Y.to_numpy().reshape(len(Y), 1)

X_train = X[:-250]
X_test = X[-250:]
Y_train = Y[:-250]
Y_test = Y[-250:]
plt.scatter(X_test, Y_test, color='black')
plt.title('Test Data')
plt.xlabel('Size')
plt.ylabel('Price')
plt.xticks(())
plt.yticks(())
regr = linear_model.LinearRegression()
regr.fit(X_train, Y_train)

plt.plot(X_test, regr.predict(X_test), linewidth=3, color='red')
plt.savefig("regression_plot.png")
print("Plot saved as regression_plot.png")
```

Output:



Classification

- Classification is a type of supervised learning in machine learning. This means the model is trained using data with labels (answers) so it can learn and make predictions on new data . In simple terms, classification helps a machine decide which group or category something belongs to.
- For example, a spam filter learns from thousands of labeled emails to recognize whether a new email is **spam** or **not spam**. Since there are only two possible outcomes, this is called **binary classification**.

Types of Classification

- Classification problems are commonly categorized into three main types based on the number of output classes:
  - Binary Classification
  - Multiclass Classification
  - Multilabel Classification

TYPES OF CLASSIFICATION

1. Binary Classification

Classifying into two categories

Spam
Not Spam

Positive
/ Negative

Default
No Default

Default /
No Default

2. Multiclass Classification

Involves more than two classes

Digit
Recognition

Sentiment
Analysis

Animal
Classification

Dog,
Bird

3. Multilabel Classification

Each instance has multiple classes

Tagging

Image
Tagging

Music
Genre

Image
Tagging

1. Binary Classification

- This involves classifying data into two categories or classes. Examples include:
- Email spam detection (Spam/Not Spam)
- Disease diagnosis (Positive/Negative)
- Credit risk prediction (Default/No Default)

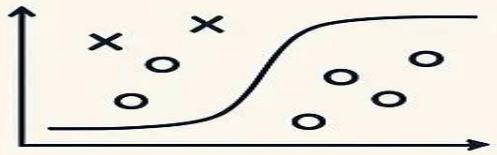
2. Multiclass Classification

- Involves more than two classes. Each input is assigned to one of several possible categories. Examples:
- Digit recognition (0–9)
- Sentiment analysis (Positive, Negative, Neutral)
- Animal classification (Cat, Dog, Bird, etc.)

3. Multilabel Classification

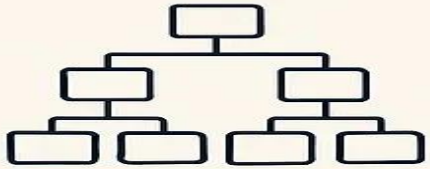
- Here, each instance can belong to multiple classes at the same time. Examples:
- Tagging a blog post with multiple topics
- Music genre classification
- Image tagging (e.g., an image may include a beach, people, and a sunset).

Popular Classification Algorithms in Machine Learning



Logistic Regression

Despite the name. logistic regre-ssion is a classification algorithm, not a regression one

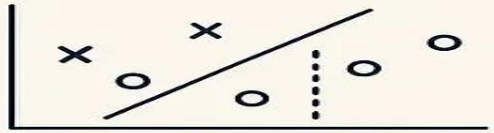


Decision Trees

Decision trees are flowchart-like structures that make decisions based on feature values

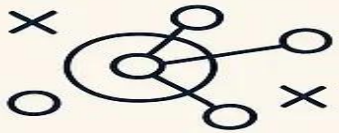
Random Forest

Random Forest is an ensemible learning method. meaning it builds not just one but many decision trees during training



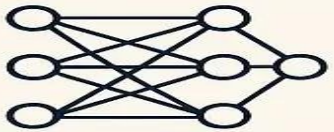
Support Vector Machines (SVM)

Support Vector Machines (SVM) is a powerful algorithm that tries to find the best boundary (hyper-plane) that separates the data points of different classes



Naive Bayes

This is a probabilistic classifier based on Bayes Theorem, which calculates the probability that a data point belongs to a particular class



Neural Networks

Neural networks are the foundation of deep learning. Inspired by the human brain, they consist of layers of interconnected nodes (neurons)

1. Logistic Regression

- Despite the name, logistic regression is a classification algorithm, not a regression one. It's commonly used for binary classification problems and outputs a probability score that maps to a class label.

Code:

```
from sklearn.linear_model
import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
```

2. Decision Trees

- Decision trees are flowchart-like structures that make decisions based on feature values. They're intuitive and easy to visualize.

Code:

```
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
```

3. Random Forest

- **Random Forest** is an **ensemble learning method**, meaning it builds not just one but many **decision trees** during training. Each tree gives a prediction, and the final output is decided by **majority voting** (for classification) or averaging (for regression).
- It helps reduce **overfitting**, which is a common problem with individual decision trees.
- Works well even with missing data or non-linear features.
- Example use case: **loan approval prediction**, **disease diagnosis**.

4. Support Vector Machines (SVM)

- Support Vector Machines (SVM) is a powerful algorithm that tries to find the best boundary (hyperplane) that separates the data points of different classes.
- Works for both **linear** and **non-linear** classification by using a **kernel trick**.
- Very effective in **high-dimensional spaces** like text data.
- Example use case: **Face detection**, **handwriting recognition**.

6. Naive Bayes

- Naive Bayes is a **probabilistic classifier** based on **Bayes' Theorem**, which calculates the probability that a data point belongs to a particular class.
- It assumes that features are **independent**, which is rarely true in reality, but it still performs surprisingly well.
- Very fast and good for **text classification** tasks.
- Example use case: **Spam filtering**, **sentiment analysis**.

7. Neural Networks

- Neural networks are the foundation of **deep learning**. Inspired by the human brain, they consist of **layers of interconnected nodes (neurons)**.
- They can model complex relationships in large datasets.
- Especially useful for **image, video, audio, and natural language data**.
- It requires more data and computing power than other algorithms.
- Example use case: **Image recognition**, **speech-to-text**, **language translation**.

Classification Examples

Example 1: Email Spam Detection

Email Text	Label
"Win a free iPhone now!"	Spam
"Your invoice for last month is here."	Not Spam

Example 2: Disease Prediction

Features	Label
Fever, Cough, Shortness of Breath	COVID-19
Headache, Sneezing, Runny Nose	Common Cold

Classifier Performance Metrics

- To evaluate the performance of a **classifier in machine learning**, the following metrics are commonly used:
- **Accuracy**: Overall correctness
- **Precision**: Correct positive predictions
- **Recall**: True positives identified
- **F1 Score**: Harmonic mean of precision and recall
- **Confusion Matrix**: Tabular view of predictions vs actuals

Choosing the Right Classification Algorithm

- When selecting a **classification algorithm**, consider the following:
- Size and quality of the dataset
- Linear vs non-linear decision boundaries
- Interpretability vs accuracy
- Training time and computational complexity
- Use cross-validation and hyperparameter tuning to optimize model performance.

Examples of Machine Learning Classification in Real Life

- Classification algorithms are widely used in many real-world applications across various domains, including:
- **Email spam filtering:** Filters our emails into spam or not spam based on email content.
- **Credit risk assessment:** Algorithms predict whether a loan applicant is likely to default by analyzing factors such as credit score, income and loan history. This helps banks make informed lending decisions and minimize financial risk.
- **Medical diagnosis**: Machine learning models classify whether a patient has a certain condition like cancer or diabetes based on medical data such as test results, symptoms and patient history. This aids doctors in making quicker, more accurate diagnoses, improving patient care.
- **Image classification :** Applied in fields such as facial recognition, autonomous driving and medical imaging.
- **Sentiment analysis:** Determining whether the sentiment of a piece of text is positive, negative or neutral. Businesses use this to understand customer opinions, helping to improve products and services.
- **Fraud detection :** Algorithms detect fraudulent activities by analyzing transaction patterns and identifying anomalies crucial in protecting against credit card fraud and other financial crimes.
- **Recommendation systems :** Used to recommend products or content based on past user behavior, such as suggesting movies on Netflix or products on Amazon. This personalization boosts user satisfaction and sales for businesses.

Applications of Supervised learning

- It can be used to solve variety of problems which includes:
- **Image classification:** It can automatically classify images into different categories such as animals, objects or scenes helps in the tasks like image search, content moderation and image-based product recommendations.
- **Medical diagnosis:** It can assist in medical diagnosis by analyzing patient data such as medical images, test results and patient history to identify patterns that suggest specific diseases or conditions.
- **Fraud detection:** They can analyze financial transactions and identify patterns that shows fraudulent activity which helps financial institutions prevent fraud and protect their customers.
- **Natural language processing (NLP):** It plays a important role in NLP tasks including sentiment analysis, machine translation and text summarization which enables machines to understand and process human language effectively.

Advantages of Supervised learning

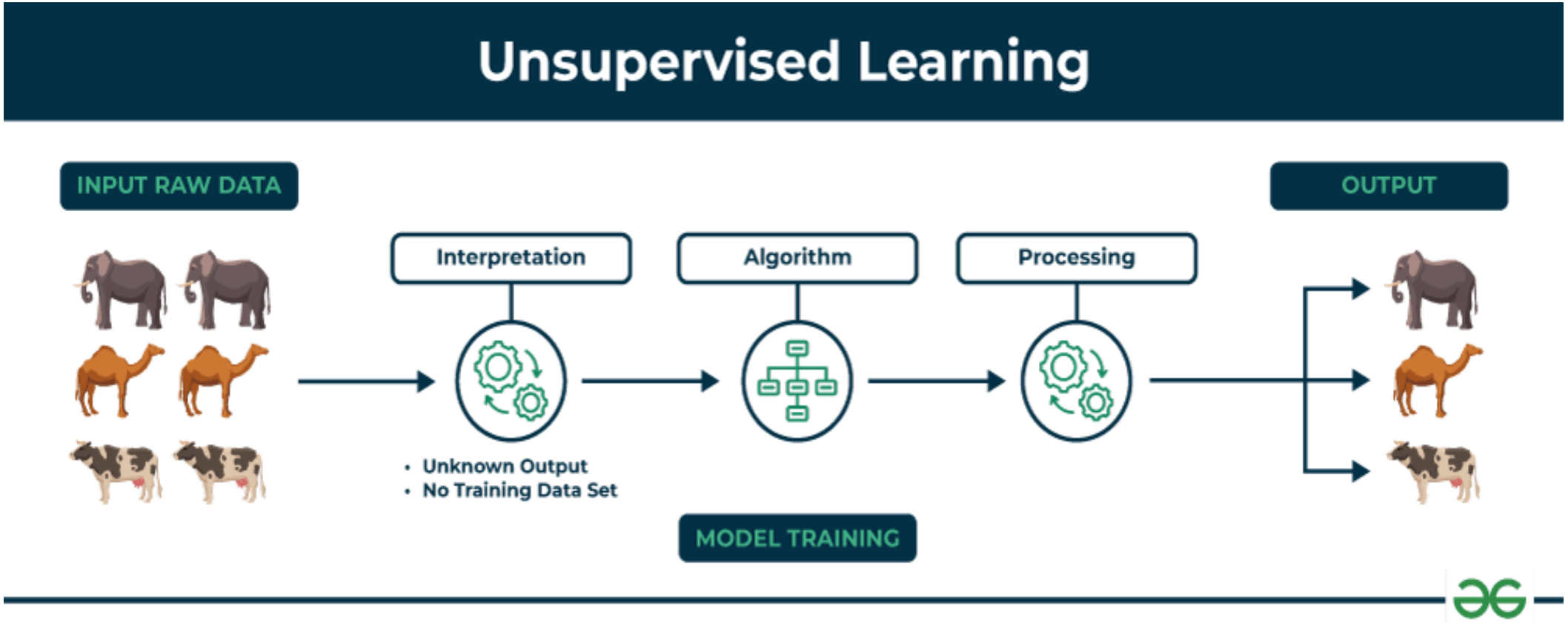
- It learns from labeled examples to make accurate predictions on new, unseen data.
- With more data and training, these models increases their accuracy which leads to better performance and more reliable predictions.
- It works well for many tasks from detecting spam emails to predicting house prices as it has the ability to handle various computational challenges.
- It can handle both classification (sorting data into categories) and regression (predicting numbers) which makes it flexible for different problems.

Disadvantages of Supervised learning

- It requires a well-labeled dataset where each input has a corresponding output. Creating such datasets takes a lot of time, money and effort and can sometimes have mistakes, this makes supervised learning hard to use.
- It works well on many tasks but can struggle with very complex or unstructured problems like understanding patterns or abstract ideas that doesn't relate to what it was trained on.
- These models can sometimes overfit the training data which means they perform well on training data but poor on new, unseen data.
- These models often need constant updating with new labeled data to stay accurate as real-world data changes over time.

Unsupervised Learning

- Unsupervised learning is a part of machine learning which works differently from supervised because there is no teacher(supervisor) involved to guide the machine.
- In this approach the machine is given with data that has **no labels or categories**. It analyzes the data on its own to find patterns, groups or relationships without any prior knowledge.
- The machine learns by discovering hidden structures within the data without being told what the correct output should be.
- For example, unsupervised learning can analyze animal data and group the animals by their traits and behavior. These groups might represent different species which allows the machine to organize animals without any prior labels or categories.


Unsupervised Learning
INPUT RAW DATA
OUTPUT
Interpretation
Algorithm
Processing
• Unknown Output
• No Training Data Set
MODEL TRAINING

Example to understand

- Imagine we have a machine learning model trained on many unlabeled images of dogs and cats. The model has never seen any labeled example that says “dog” or “cat” before so it doesn’t know how these animals look like.
- Now, if we give the model a new image that contains both dogs and cats it won’t be able to directly label them as “dog” or “cat.” It will group parts of the image based on similarities and differences in features like shape or texture. It might separate the image into two groups one with dog-like features and other with cat-like features.
- This happens because unsupervised learning doesn’t rely on prior knowledge or training with labeled data. It finds patterns and organizes data on its own helps in discovering information that wasn’t given before.

Types of Unsupervised Learning

- Unsupervised learning is divided into two categories of algorithms:

1. Clustering

- A clustering is used to group similar data points together. Clustering algorithms work by repeatedly moving data points closer to to the center of their group (cluster) and farther from points in other groups. This helps the algorithm to create clear and meaningful clusters. Some popular clustering algorithms include:
- K-means clustering
- Hierarchical clustering
- Principal Component Analysis (PCA)
- Singular Value Decomposition (SVD)
- Independent Component Analysis
- Gaussian Mixture Models (GMMs)
- Density-Based Spatial Clustering of Applications with Noise (DBSCAN)

2. Association rule learning

- An association rule learning used to find patterns and relationships between different items in a dataset. It looks for rules like “people who buy X often also buy Y”.
- Some common Association rule learning algorithms include:
- Apriori Algorithm
- Eclat Algorithm
- FP-Growth Algorithm

Application of Unsupervised learning

- Unsupervised learning can be used to solve a variety of problems which includes:
- **Anomaly detection**: It can identify unusual patterns or behaviors in data helps in the detection of fraud, security breaches or system problems.
- **Scientific discovery**: It can show hidden relationships and patterns in scientific data which leads to new insights and ideas.
- **Recommendation systems**: It finds similarities in user behavior and preferences to recommend products, movies or music that align with their interests.
- **Customer segmentation**: It can identify groups of customers with similar characteristics which allows businesses to target marketing campaigns and improve customer service more effectively.

Advantages of Unsupervised learning

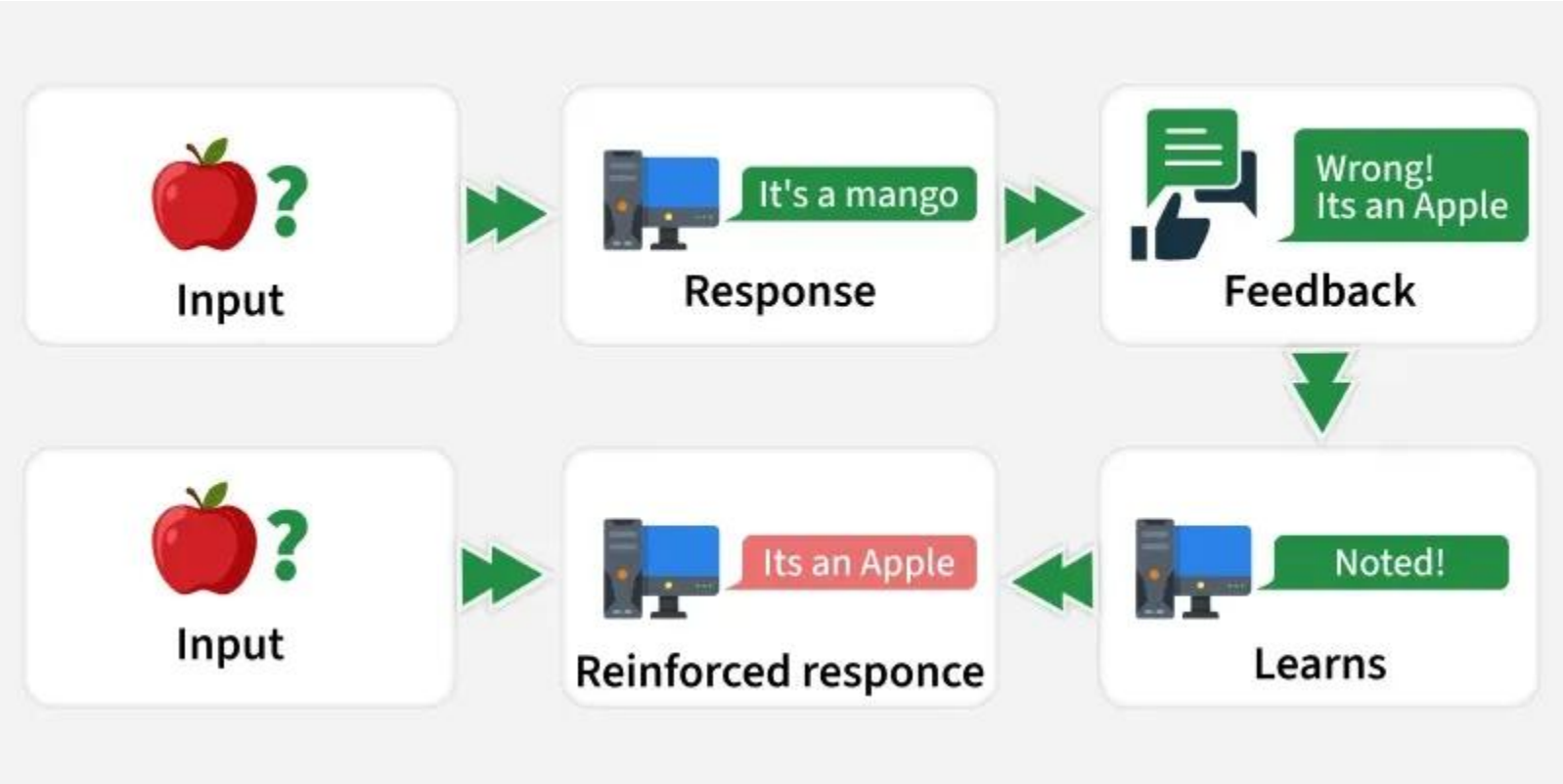
- It doesn't need labeled data so we can start working with large datasets more easily and quickly.
- This handles large amounts of data and reduce it into simpler forms without losing important patterns which makes it manageable and efficient.
- It discovers patterns and relationships in the data that were previously unknown which offers valuable insights.
- By analyzing unlabeled data, it shows meaningful trends and groups that help us to understand our data deeply.

Disadvantages of Unsupervised learning

- Without labeled answers, it's difficult to tell how accurate or effective the model is.
- Lack of clear guidance can lead to less precise results for complex problems.
- After grouping the data, we may needs to check and label these groupings which can be time-consuming.
- Missing data, outliers or noise in the data can easily affect the quality of the results.

Reinforcement Learning

- Reinforcement Learning (RL) trains an agent to make decisions by interacting with an environment. Instead of being told the correct answers, agent learns by trial and error method and gets rewards for good actions and penalties for bad ones. Over time it develops a strategy to maximize rewards and achieve goals. This approach is good for problems having sequential decision making such as robotics, gaming and autonomous systems.
- Example: While Identifying a Fruit, system receives an input for example an apple and initially makes an incorrect prediction like "It's a mango". Feedback is provided to correct the error "Wrong! It's an apple" and the system updates its model based on this feedback.
- Over time it learns to respond correctly that "It's an apple" when getting similar inputs and also improves accuracy.


Input
It's a mango
Response
Wrong!
Its an Apple
Feedback
Noted!
Learns
Its an Apple
Reinforced responce
Input

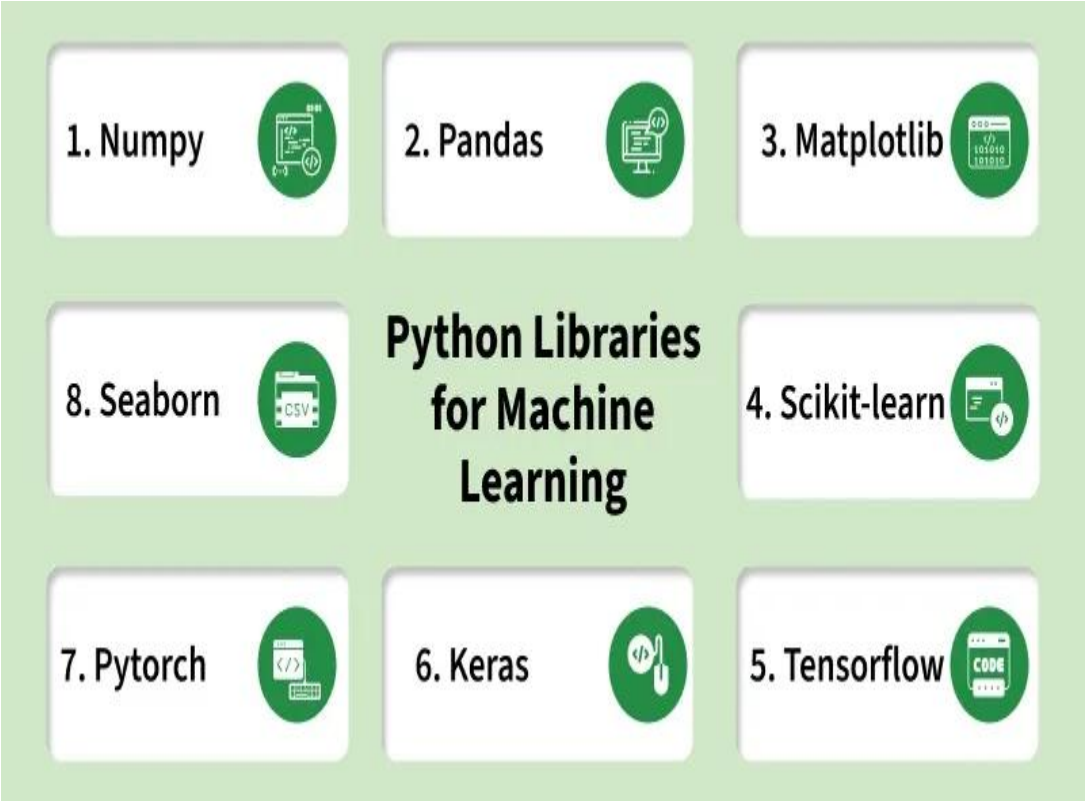
Supervised vs Unsupervised Machine Learning

Parameters	Supervised machine learning	Unsupervised machine learning
Input Data	They are trained on labeled data.	They are trained on unlabeled data.
Computational Complexity	Simpler method	Computationally complex
Accuracy	Highly accurate	Less accurate
No. of classes	No. of classes is known	No. of classes is not known
Data Analysis	Uses offline analysis	Uses real-time analysis of data
Algorithms used	Linear and Logistics regression, KNN Random forest, multi-class classification, decision tree, Support Vector Machine, Neural Network etc.	K-Means clustering, Hierarchical clustering, Apriori algorithm etc.
Output	Desired output is given.	Desired output is not given.
Training data	Use training data to infer model.	No training data is used.

Complex model	It is not possible to learn larger and more complex models with supervised learning.	It is possible to learn larger and more complex models with unsupervised learning.
Model	We can test our model.	We can not test our model.
Supervision	Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Classification	Divided into two types: 1.Regression 2.Classification	Divided into two types: 1.Clustering 2.Association
Feedback	It has feedback mechanism.	It has no feedback mechanism.
Time Consumption	It's more time consuming.	It's less time consuming.
Example	Optical character recognition.	Find a face in an image.

Python libraries for Machine Learning

- Machine Learning involves building systems that can automatically learn patterns from data and make predictions or decisions without explicit programming. Python has emerged as the most widely used language for machine learning due to its simplicity, readability and its useful ecosystem of libraries. These libraries provide efficient tools for data handling, visualization, feature engineering, model building and evaluation making the entire machine learning workflow faster and more reliable.
- They provide optimized implementations of complex algorithms
- They simplify data preprocessing and feature engineering
- They support rapid experimentation and prototyping
- They are widely used in both academia and industry
- Some popular Python libraries for Machine Learning are:



1. NumPy

- NumPy is a fundamental numerical computing library in Python that provides support for large, multi-dimensional arrays and matrices, along with a comprehensive collection of mathematical functions. In machine learning, NumPy is primarily used for handling numerical data, performing vectorized operations and implementing low-level mathematical computations efficiently.
- Used for numerical feature representation and transformation
- Enables fast mathematical operations through vectorization
- Serves as the computational backbone for many ML libraries
- Efficient memory management for large datasets

2. Pandas

- Pandas is a high-level data analysis and manipulation library built on top of NumPy. It introduces useful data structures such as DataFrame and Series, which allow machine learning practitioners to clean, transform and analyze structured data efficiently before feeding it into models.
- Used for data cleaning, transformation and preparation
- Handles missing, inconsistent and categorical data
- Simplifies exploratory data analysis
- Integrates seamlessly with ML and visualization libraries

3. Matplotlib

- Matplotlib is a comprehensive data visualization library used to create static and interactive plots. In machine learning, it plays a critical role in understanding data distributions, detecting patterns and interpreting model performance through graphical representations.
- Used for visualizing datasets and model outputs
- Helps identify trends, skewness and imbalances
- Supports custom and publication-quality plots
- Essential for result interpretation

4. Scikit-learn

- Scikit-learn is a widely used machine learning library that provides simple and efficient tools for classical machine learning tasks. It supports supervised and unsupervised learning algorithms along with preprocessing, model evaluation and validation utilities.
- Used for classification, regression and clustering
- Provides consistent and easy-to-use API
- Includes preprocessing and evaluation tools
- Ideal for traditional ML problems

5. TensorFlow

- TensorFlow is a useful open-source deep learning framework developed by Google. It is designed for building, training and deploying large-scale neural networks and supports both research and production-level machine learning systems.
- Used for deep learning and neural networks
- Supports GPU and distributed training
- Highly scalable and production-ready
- Flexible model architecture design

6. Keras

- Keras is a high-level neural network API that simplifies deep learning model development. It abstracts much of the complexity involved in building neural networks, making it especially suitable for beginners and rapid prototyping.
- Simplifies neural network creation
- Requires minimal code
- Supports both regression and classification
- Improves development speed

7. PyTorch

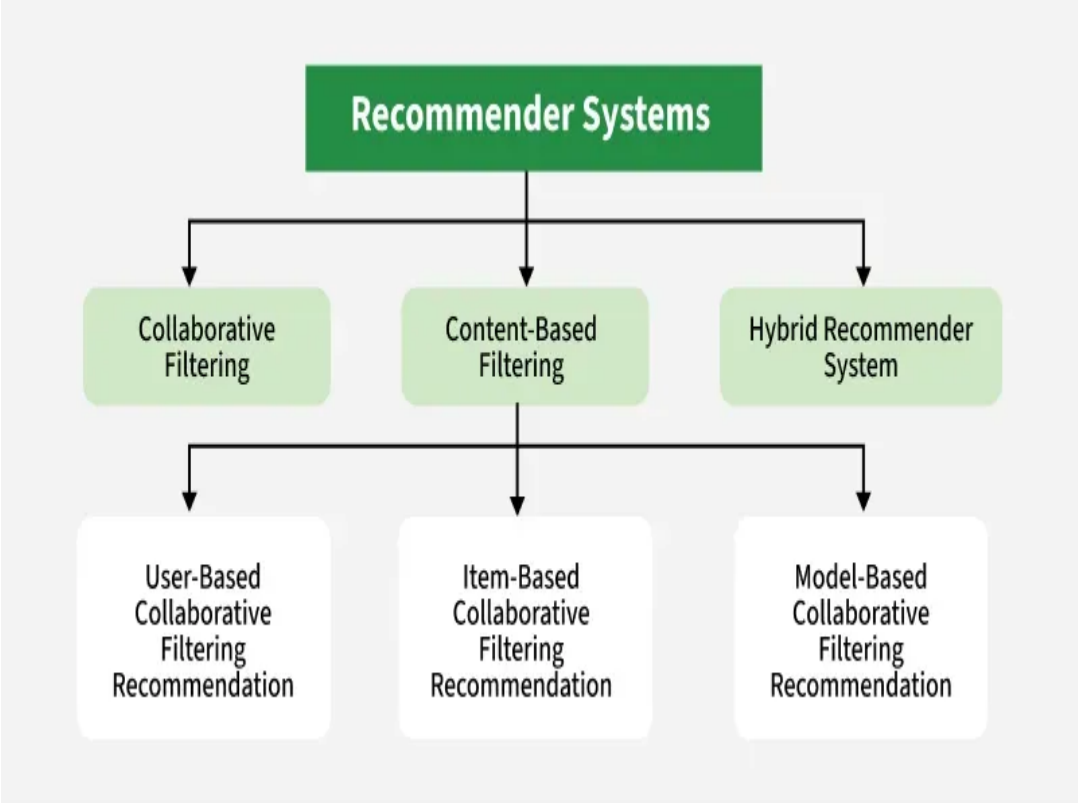
- PyTorch is an open-source deep learning library known for its dynamic computation graph, which allows models to be modified during execution. This makes PyTorch highly flexible and popular in research and experimentation.
- Used for research-oriented deep learning
- Dynamic and intuitive model building
- Easier debugging and customization
- Supports custom training logic

8. Seaborn

- Seaborn is a statistical data visualization library built on Matplotlib. It is designed to create informative and visually appealing plots that help in understanding relationships between variables during exploratory data analysis.
- Used for exploratory data analysis
- Works directly with pandas DataFrames
- Produces cleaner statistical plots
- Enhances data interpretation

Recommender Systems

- Recommender Systems are tools that suggest items to users based on their behaviour, preferences or past interactions. They help users find relevant products, movies, songs or content without manually searching for them. They are widely used in platforms like YouTube, Amazon and Netflix.
- Suggest items automatically
- Use user behavior and patterns
- Improve content discovery
- Common in e-commerce and entertainment apps



Types of Recommender Systems

Let's see the various types:

1. Content-Based Filtering: Content-based filtering recommends items similar to those a user liked earlier by analyzing item features and user preference profiles.

•Useful when users have strong, consistent preferences.

•Seen in movie apps suggesting films of the same genre or style.

•**Advantage**: Personalization without needing other users data.

•**Limitation**: Lack of novelty because it focuses on similar items only.

2. Collaborative Filtering: Collaborative filtering identifies similarities in user behavior and recommends items based on patterns derived from many users.

•Works well with large datasets of user interactions.

•Used in Amazon's "Users who bought this also bought" and Netflix suggestions.

•**Advantage**: Discovery of new items outside a user's past history.

•**Limitation**: Cold-start issues with new items or new users.

3. Hybrid Recommendation Systems: Hybrid systems combine collaborative and content-based methods to offer more accurate and robust recommendations.

•Useful where both user behavior and item features provide value.

•Seen in platforms like Netflix that blend metadata and viewing patterns.

•**Advantage**: Reduced weaknesses and better overall performance.

•**Limitation**: Increased complexity and higher computational cost.

4. Knowledge-Based Recommendation Systems: Knowledge-based systems use explicit domain knowledge and user requirements rather than historical behavior to suggest items.

•Ideal for high-involvement domains like real estate or financial products.

•Seen in travel planning apps or loan product matchers.

•**Advantage**: No need for past interactions.

•**Limitation**: Effort required to build domain-specific rules.

5. Context-Aware Recommendation Systems: These systems use contextual information such as time, location, device or mood to personalize recommendations.

•Effective in mobile and location-based applications.

•Seen in restaurant suggestions based on time of day or location.

•**Advantage**: Higher relevance through situational awareness.

•**Limitation:** Complexity in capturing and processing contextual data.

Working

- Let's see how a recommender system works,
- **Data Collection:** Collect ratings, clicks, views, purchases, searches and item metadata.
- **Data Preprocessing:** Clean data, handle missing values and prepare user–item matrices.
- **Feature Engineering:** Transform user and item characteristics into meaningful vectors.
- **Model Training / Similarity Computation:** Use algorithms like collaborative filtering, content-based models or deep learning.
- **Prediction:** Generate relevance scores for unseen items based on learned patterns.
- **Ranking & Recommendation:** Sort items by predicted relevance and recommend the top ones.
- **Feedback Loop:** Update the model continuously as new user interactions arrive

Deep Learning Models Used in Recommender Systems

- Let's see the various types of deep learning models that are used in recommender systems,

1. Neural Collaborative Filtering (NCF)

- Learns non-linear and complex user–item interaction patterns using multilayer neural networks.
- Offers better flexibility than traditional matrix factorization by learning interactions directly from data.

2. Autoencoders

- Encode sparse user–item matrices into dense latent embeddings and reconstruct missing values.
- Useful for denoising, dimensionality reduction and generating more accurate rating predictions.

3. Recurrent Neural Networks (RNNs)

- Model sequential user behavior by capturing the order of interactions (e.g., the next movie or product a user is likely to consume).
- Effective for session-based and time-dependent recommendation tasks.

4. Convolutional Neural Networks (CNNs)

- Extract deep feature representations from images, text or audio to support content-based recommendations.
- Common in scenarios like recommending visually similar products or tagging video content.

5. Transformer Models

- Use attention mechanisms to understand long-range dependencies in user interaction sequences.
- Provide state-of-the-art performance in next-item prediction and context-aware recommendation.

6. Wide & Deep Models / DeepFM

- Combine memorization (feature interactions) and generalization (deep neural networks) for large-scale recommendation problems.
- Widely adopted in advertising and e-commerce platforms due to strong performance with structured data.

Importance

- Let's see the importance of recommender systems,
- Improve user satisfaction by providing relevant and personalized recommendations.
- Reduce information overload by efficiently filtering large volumes of content.
- Increase user engagement by helping users discover items they are likely to enjoy.
- Boost conversions and revenue through targeted and data-driven suggestions.
- Enhance customer retention by delivering consistent, personalized interactions.

Benefits of Machine Learning

- **Enhanced Efficiency and Automation:** ML automates repetitive tasks, freeing up human resources for more complex work. This leads to faster, smoother processes and higher productivity.
- **Data-Driven Insights:** It can analyze large amounts of data to identify patterns and trends that might be missed by people and help businesses make better decisions.
- **Improved Personalization:** It customizes user experiences by tailoring recommendations and ads based on individual preferences.
- **Advanced Automation and Robotics:** It helps robots and machines to perform complex tasks with greater accuracy and adaptability. This is transforming industries like manufacturing and logistics.

Challenges of Machine Learning

- **Data Bias and Fairness:** ML models learn from training data and if the data is biased, model's decisions can be unfair so it's important to select and monitor data carefully.
- **Security and Privacy Concerns:** Since it depends on large amounts of data, there is a risk of sensitive information being exposed so protecting privacy is important.
- **Interpretability and Explainability:** Complex ML models can be difficult to understand which makes it difficult to explain why they make certain decisions. This can affect trust and accountability.
- **Job Displacement and Automation:** Automation may replace some jobs so retraining and helping workers learn new skills is important to adapt to these changes.

Applications of Machine Learning

- Machine Learning is used in many industries to solve problems and improve services. Here are some common real-world applications:
- **Healthcare:** It helps doctors to diagnose diseases from medical images like X-rays and MRIs. It also predicts patient outcomes and personalizes treatments which improves healthcare quality.
- **Finance:** In finance it detects fraudulent transactions in real time and supports algorithmic trading. It also helps to assess credit risk helps in making lending safer and faster.
- **Retail and E-Commerce:** It helps in personalized product recommendations and forecasts demand to optimize inventory and also analyzes customer sentiment to improve shopping experiences.
- **Transportation and Automotive:** Self-driving cars rely on ML to navigate and make decisions. It optimizes delivery routes and predicts vehicle maintenance needs which reduces downtime.
- **Social Media and Entertainment:** Platforms like Netflix and YouTube use ML to recommend content we'll enjoy. It enables image and speech recognition for better user interaction.
- **Manufacturing:** It improves quality control by detecting defects in products automatically and predicts machine failures in advance and helps in production processes.

Thank You....